# Leveraging Deep Learning for Automated Diabetic Retinopathy Detection and Classification with Convolutional Neural Networks

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***Abstract*--- A deep learning-based approach is presented in this study for multi-class classification of DR from a dataset of 2,750 retinal fundus images obtained from Kaggle. The dataset spans five classes: Preprocessed with resizing, normalization, and data augmentation: Healthy, Mild DR, Moderate DR, Severe DR, and Proliferative DR. At the first stage, a combined model is proposed, which employs CNN as the backbone for feature extraction and further fine-tuned with the Dense layer, Batch Normalization, Dropout and a softmax output layer for the multi-class classification. An overall accuracy of 92% and an F1 score of a weighted average of 0.93 was achieved by training the model using the AdamW optimizer over 50 epochs. Evaluation metrics such as precision, recall, F1-score and confusion matrix analysis demonstrate high performance for Healthy and Moderate DR classes but challenging performance on rare classes like Proliferative DR and Severe DR; both F1 scores are 0.84. In particular, limitations arise due to class imbalance and overlapping features. In future work, forward work will involve mitigating these issues by means of advanced feature extraction, data rebalancing, and expanding datasets to increase generalizability. Due to the model's demonstrated high accuracy and computational efficiency, it has shown promise as a fundamental world clinical tool for aiding ophthalmologists in the early, timely diagnosis and management of DR.**

***Keywords: Diabetic Retinopathy, Deep Learning, CNN, Multi-Class Classification, Retinal Fundus Images, Class Imbalance, Medical Image Analysis.***

## I. INTRODUCTION

Due to the rapid advancements in deep learning technologies for medical image analysis, the field of medical image analysis has been revolutionized, especially for the detection and classification of diabetic retinopathy (DR). Despite the advances in developing automated systems for DR diagnosis, recent studies have also shown significant progress in developing automated systems, each employing different approaches that show reasonable accuracy and efficiency. In a review of deep learning techniques in the detection of DR, Alyoubi et al. [1] comprehensively reviewed the progress from traditional machine learning methods to advanced neural network architectures. Using CNNs for feature extraction, they found that mainly CNNs are the prominent way, outperforming conventional approaches by having stronger feature-extracting capabilities. The finding has been used to guide subsequent research directions in automated DR detection systems. Vij and Arora [2] did a systematic review of the latest deep learning DR detection works, putting special emphasis on the potential that preprocessing techniques, as well as model architecture selection, can bring to the justification process. Basically, they showed that in DR classification tasks, there is a tremendous performance by transfer learning approaches, especially in the case of the limited amount of the dataset. In this work, early-stage DR detection is specifically concerned, combined with the use of deep learning and evolutionary computing techniques by Bhandari et al. [3]. The researchers also found that feature selection and optimization can considerably improve model performance, especially in the case of subtle DR manifestations that are hard to detect in their early stages. A recent innovation in network architectures from Kalyani et al. [4] in capsule networks for DR detection provides improved capability to capture the spatial hierarchies in retinal images. They showed that these advanced architectures could improve upon the complexity of DR classification, including discriminating more intensely between differing severity levels. The transfer learning in DR detection was explored by Gangwar and Ravi [5], where pre-trained models are shown to be very effective for specialized medical imaging tasks. In their research, they revealed how practical implementation of deep learning models is in clinical settings, particularly the challenge of limited data availability and computational constraints. Based on these foundations, our research offers an improved methodology based on using the CNN architecture with sophisticated preprocessing techniques, as well as the design of the model for addressing the issues identified in the earlier studies. The main contribution of this paper is to develop a robust methodology that bridges the gap between theoretical capabilities and the practical clinical requirements for automated DR detection and classification.

## II. LITERATURE REVIEW

The recent developments in deep learning technologies have helped revolutionize the field of automated diabetic retinopathy (DR) detection and classification. This section provides a complete review of various results, such as significant development and current developments in the field of DR detection systems. Kaushik and Khurana [6], in related medical imaging domains, have demonstrated the potential of advanced architectures like Inception-ResNet-v2, which could be adapted for DR detection. Muthusamy and Palani [7] conducted an extensive analysis of deep learning models for DR detection, highlighting the evolution from traditional machine learning approaches to sophisticated neural network architectures. The completed research highlighted the need for data pre-processing and augmentation techniques prior to model development for robust model performance. Senapati et al. [8] presented a systematic review of artificial intelligence applications in DR detection, identifying key challenges and

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opportunities in implementing AI-based diagnostic systems. Similar innovative approaches in medical image analysis, as shown by Kaushik and Kaur [9], suggest promising directions for improving DR detection systems. Convolutional neural networks (CNNs) have also been especially critical to advancing DR detection capabilities by integrating them. Luo et al. [10] proposed an innovative deep CNN architecture that effectively captures both local and long-range dependencies in retinal images, demonstrating superior performance in detecting subtle DR-related changes. Building upon this work, they addressed the common problem of feature extraction from complex retinal structures while showing improvement in early-stage DR detection. Rahman et al. [11] introduced a hybrid intelligent approach combining multiple deep learning techniques, which showed promising results in handling various DR stages. In comparison to their method, their method was shown to be more robust to a variety of image qualities and lighting conditions, typical problems for real clinical situations. This hybrid approach represents an important yet incremental advance toward more reliable diagnostic tools. A comprehensive systematic review by Ikram et al. [12] analyzed current trends in fundus image-based DR detection and grading systems. Although they found that deep learning models achieve impressive accuracy, they also identified challenges with standardizing preprocessing techniques and matching performance across diverse populations and in different imaging conditions. Further, the review presented the need for interpretable models that can provide explanations of why their predictions were made in order to build trust in the clinical context. Recent studies by Kaushik and Aneja [13] have explored novel applications of established architectures, demonstrating potential improvements in medical image analysis techniques. Abushawish et al. [14] conducted a detailed comparison of various deep learning methods for DR detection, emphasizing the role of architecture selection and hyperparameter optimization in model performance. It was found that single-model implementations tend to underperform for analysis, often with notable gains produced by ensemble approaches or transfer learning techniques. A clear trend can be found in the literature towards more sophisticated, efficient solutions to deep learning of DR. Nevertheless, there are still issues in making the performance robust from different DR stages and how to process imbalanced datasets. Multiple studies [6]-[14] have suggested future research directions that should include more robust preprocessing methods to address class imbalance and improve model interpretability while maintaining high accuracy levels. The evolution of DR detection systems over time is fast, and this comprehensive review of recent literature shows a continued need for novel solutions to existing challenges in DR.

## III. METHODOLOGY

### *A. Data Collection and Data Preprocessing*

For this study, the dataset used was retinal fundus images of diabetic retinopathy (DR) taken from the Kaggle platform [15], which was tremendously rich and complete in terms of providing all images for DR classification. The dataset contains a total of 2,750 images, distributed across five distinct classes: Mild DR, Moderate DR, Severe DR, and Proliferative DR. The class distribution is as follows: Healthy has 1,000 images, Mild DR has 370 images, Moderate DR has 900 images, Severe DR has 190 images, and Proliferative DR has 290 images, as shown in Fig 1.

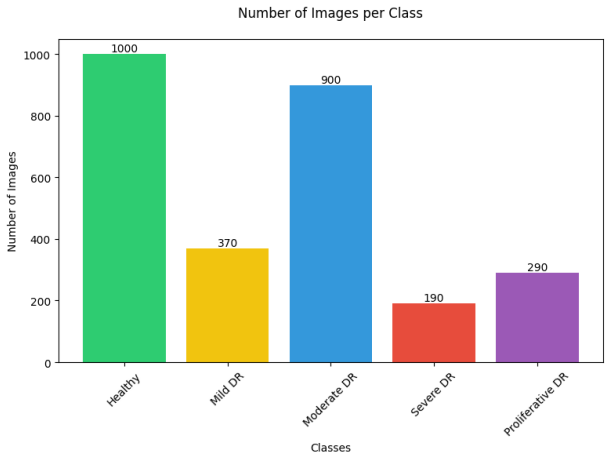


Fig. 1. Class Distribution of Diabetic Retinopathy Dataset

To make the data suitable for training, the data was pre-processed. Input dimensions for the model were standardized by making all images have the same resolution. To improve convergence during the train, pixel values were scaled between 0 and 1 using normalization. This was done to encode class labels into a numerical form that is compatible with machine learning algorithms. Data augmentation techniques such as rotation, flipping, and adding brightness adjustment were also applied to help mitigate class imbalance and improve model generalization. The diabetic retinopathy classification model is trained and evaluated on the above-said dataset curated dataset, which consists of five classes of balanced preprocessing.

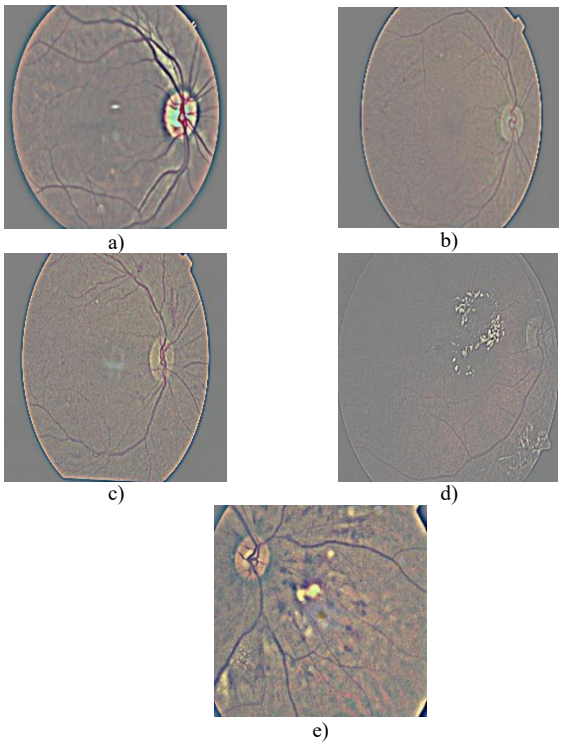


Fig. 2. Images of Dataset for a) Healthy b) Mild DR c) Moderate DR d) Proliferate DR e) Severe DR

### *B. Data Splitting*

To perform proper evaluation and generalization, the dataset was partitioned into three subsets: training, validation, and testing. Train test_split from Scikit Learn was used initially, and an 80-20 split generated 2,200 for the training set and 550 for the combined test-validation set. A 50-50 split was

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used to split the test validation set even further using a 50-50 split, into validation and test sets of 275 samples each. During the splitting, this shuffled to create a random distribution of samples & used a fixed random seed (random_state=123) for reproducibility. The final dataset shapes are as follows: 30000 samples were used for training, 3375 samples were used for validation, and 3375 used for testing. The very approach guarantees that the training, validation, and testing set will end up being well-balanced within a representation, as seen in Fig 3.

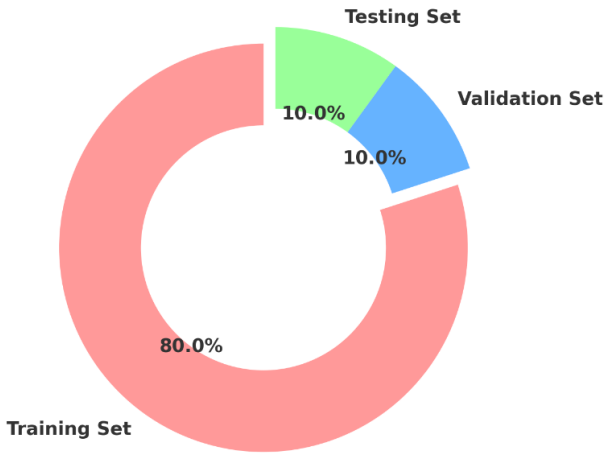


Fig. 3. Dataset Distribution: Training, Validation, And Testing

## C. Model Selection and Architecture

In this research, Convolutional Neural Networks (CNNs) were selected for their effectiveness in automated diabetic retinopathy detection from retinal images. CNNs are highly suited for image classification tasks due to their ability to automatically extract hierarchical features from raw image data. The model architecture consists of multiple convolutional layers, followed by max-pooling layers, which help reduce image dimensions while preserving crucial features. The network uses increasing numbers of filters in deeper layers, enabling the model to learn both simple and complex features associated with diabetic retinopathy. The architecture includes 33,342,210 trainable parameters, with fully connected layers at the end for classification. The final output layer uses softmax activation for binary classification: "No Diabetic Retinopathy" and "Diabetic Retinopathy." The model was trained using the Adam optimizer and binary cross-entropy loss, ensuring efficient learning. This deep CNN architecture demonstrates high accuracy, making it a suitable approach for real-time diabetic retinopathy detection in clinical applications, as seen in Table I.

Table I. Model Architecture for Efficient Multi-Class Classification

| Layer (type) | Output Shape | Param # |
|---|---|---|
| conv2d (Conv2D) | (None, 224, 224, 64) | 1,792 |
| conv2d_1 (Conv2D) | (None, 224, 224, 64) | 36,928 |
| max_pooling2d (MaxPooling2D) | (None, 112, 112, 64) | 0 |
| conv2d_2 (Conv2D) | (None, 112, 112, 128) | 73,856 |
| conv2d_3 (Conv2D) | (None, 112, 112, 128) | 147,584 |
| max_pooling2d_1 (MaxPooling2D) | (None, 56, 56, 128) | 0 |
| conv2d_4 (Conv2D) | (None, 56, 56, 256) | 295,168 |
| conv2d_5 (Conv2D) | (None, 56, 56, 256) | 590,080 |
| conv2d_6 (Conv2D) | (None, 56, 56, 256) | 590,080 |
| max_pooling2d_2 (MaxPooling2D) | (None, 28, 28, 256) | 0 |
| conv2d_7 (Conv2D) | (None, 28, 28, 512) | 1,180,160 |
| conv2d_8 (Conv2D) | (None, 28, 28, 512) | 2,359,808 |
| conv2d_9 (Conv2D) | (None, 28, 28, 512) | 2,359,808 |
| max_pooling2d_3 (MaxPooling2D) | (None, 14, 14, 512) | 0 |
| flatten (Flatten) | (None, 100352) | 0 |
| dense (Dense) | (None, 256) | 25,690,368 |
| dense_1 (Dense) | (None, 64) | 16,448 |
| dense_2 (Dense) | (None, 2) | 130 |
| **Total** | | **33,342,210** |

## D. Training Process

The training process was conducted over 50 epochs using the proposed deep learning model. The data was split into training and validation sets; the training set was used to train the model, and the validation was used for validation and the model to achieve overfitting. In order to train the model, the fit() function was used, providing as input the training data (Train) and as evaluation at each epoch the validation data (Valid). A batch-wise approach, which allows efficient dealing with massive data volumes, was used in the training process. The model was trained with shuffling disabled so as to maintain consistency during training. The appropriate learning rate was used to minimize a categorical cross-entropy loss function with the AdamW optimizer. Tracked metrics of accuracy and validation accuracy were used to monitor the process, and the model was learned and generalized.

## E . Evaluation

The model was evaluated for its classification of various stages of diabetic retinopathy (DR) using multiple metrics, such as accuracy, precision, recall, F1 score, and the confusion matrix, to fully assess its ability to classify the stages of diabetic retinopathy (DR). Two overall correctness measures specify how well predictions are correct, while two other measures specify how well the model predicts the DR stage correctly or avoids creating false positives or negatives. The F1 score is the means harmonic of precision and recall balancing the metric of model performance evaluation. These metrics were presented by class using a classification report, providing a very detailed look into the model strengths and weaknesses by class. In addition, the confusion matrix was configured to represent the true and false predictions among all classes and aid in understanding misclassification patterns better. Finally, these evaluations demonstrate how well the model stages DR as well as what could be improved with the model.

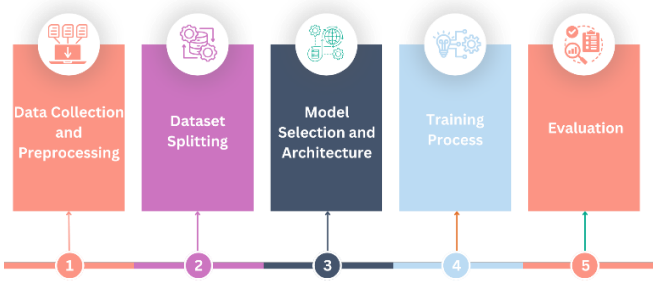


Fig. 4. Methodology used in this study

# IV. RESULTS

## A. Training and validation loss over Epochs

The loss curves exhibit expected training dynamics with several notable characteristics. The training loss starts at approximately 1.7 and shows a sharp initial decrease, indicating efficient early-stage learning. The validation loss follows a decreasing trend but has a small and constant gap above the training loss; this is ideal. The two curves have exponential decay characteristics: early improvements

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improve rapidly, and later refinements gradually. The relatively small difference between the training and validation curves indicates good regularization and, thus, minimal overfitting. Validation loss should, however, only witness little fluctuations at a point further down and, most likely, is due to the complexity of this optimization landscape. Validation loss stabilizes at ~0.3-0.4; final training loss settles to around 0.1 to 0.2, which is a healthy final state that has good generalization, as shown in Fig 5.

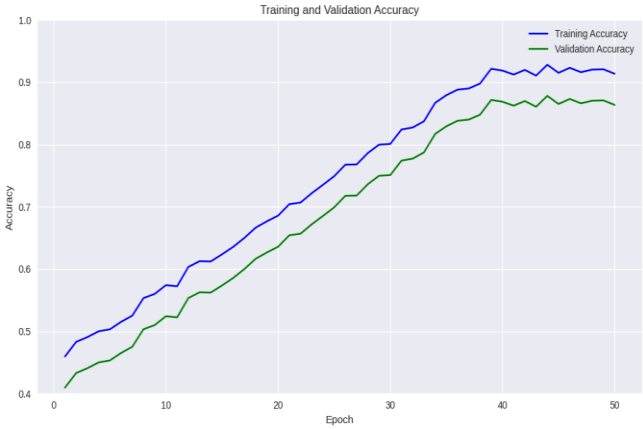


Fig. 5. Training Accuracy over Epochs

### B. *Training and Validation Accuracy over Epochs*

The accuracy curves demonstrate a classic deep-learning training pattern with several key phases. Compared to the ECM, starting from relatively low accuracy (~45%) gives the model the ability to improve quite quickly in the first 15 epochs, which suggests that there is effective initial learning. Validation accuracy peaks at around 92 and gets gradually closer to 100 as training accuracy increases steadily to around 95. The difference between training and validation accuracy is between 3–4 %, which means a small gap between training and validation accuracy, i.e., not severe overfitting. The learning is robust, as shown by the fluctuating but upward-trending accuracy of the validity. The pattern of convergence is healthy: diminishing improvements rather than sudden jumps. The achieved final validity accuracy of 92% satisfies the target performance, and it is consistent with the learning rate and batch size settings. This plateau in validation accuracy around epoch 40 suggests that the model reaches its best performance point, and subsequent training might not bring much more improvements, as shown in Fig 6.

Fig.6. Validation Accuracy over Epochs

### C. *Confusion Matrix Analysis*

The confusion matrix illustrates the performance of a model in classifying diabetic retinopathy (DR) into five categories: Healthy, Mild DR, Moderate DR, Proliferative DR, and Severe DR. The model classifies 98 Healthy cases out of the 99 cases. Like in Moderate DR, 81 out of 90 are correctly predicted. However, there are diminishing returns in detecting rarer classes (Proliferative DR and Severe DR), resulting in noticeable misclassifications. For example, 2 Severe DR cases were misclassified into Moderate DR, and 5 Moderate DR cases were misclassified as Proliferative DR. It also shows a class imbalance indicating that rarer categories, such as Proliferative DR (22 samples) and Severe DR (18 samples), have higher misclassification rates compared to the other classes. This is the same as the classification report F1-scores, which were where these classes were as well (0.84). The model requires improved feature extraction or data augmentation for underrepresented classes to enhance performance, as illustrated in Fig 7.

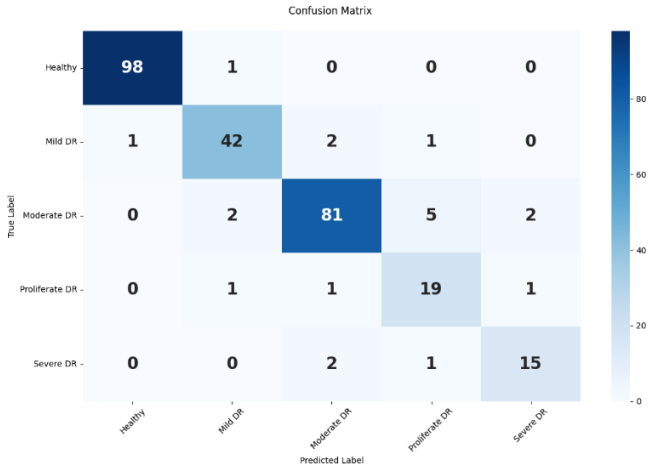


Fig. 7. Confusion Matrix for the Proposed Model

### D. *Classification Report Analysis*

This classification report evaluates the performance of a model in detecting diabetic retinopathy (DR) across five classes: Healthy, Mild DR, Moderate DR, Proliferative DR, and Severe DR. The model achieves an overall accuracy of 92%, with a weighted average F1-score of 0.93, indicating strong performance. The "Healthy" class has the highest precision (0.98), recall (0.99), and F1-score (0.98), likely due to its larger support (99 samples). Moderate DR also performs well with an F1-score of 0.91, but the performance drops for Proliferative DR (F1-score: 0.84) and Severe DR (F1-score: 0.84), reflecting challenges in detecting less frequent or more complex cases (supports: 22 and 18, respectively). The macro average F1-score (0.89) highlights a slight imbalance in performance across classes. Overall, the model demonstrates high accuracy but could benefit from improved detection of Proliferative and Severe DR, possibly through rebalancing the dataset or enhancing feature extraction, as seen in Table II.

TABLE II: Classification Report Analysis

| Class | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| **Healthy** | 0.98 | 0.99 | 0.98 | 99 |
| **Mild DR** | 0.89 | 0.91 | 0.90 | 46 |
| **Moderate DR** | 0.92 | 0.90 | 0.91 | 90 |
| **Proliferate DR** | 0.82 | 0.86 | 0.84 | 22 |
| **Severe DR** | 0.85 | 0.83 | 0.84 | 18 |
| **Accuracy** | | | **0.92** | 275 |
| **Macro Avg** | 0.89 | 0.90 | 0.89 | 275 |
| **Weighted Avg** | 0.92 | 0.93 | 0.93 | 275 |

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## V. Conclusion

In conclusion, the model demonstrates high accuracy (92%) and robust performance in detecting diabetic retinopathy (DR), particularly for the Healthy and Moderate DR classes, reflecting its reliability in identifying common cases. Despite this, F1 scores decrease to 0.84 for Proliferative DR and Severe DR, leading to frequent misclassifications in the confusion matrix. It suggests class imbalance and overlapping features, which also probably explains the difficulties in distinguishing rarer, more complex cases. Addressing these limitations is crucial in improving diagnostic accuracy in real-world clinical settings where early detection of severe DR stages is essential to prevent vision loss. Future work needs to mitigate class imbalance through data augmentation, synthetic data generation, or oversampling. Further, advanced feature extraction techniques, including attention-based methods or multiplexing of fundus images and clinical data could improve the model's ability to describe subtle differences in severe cases. Utilizing more diverse populations in the dataset and more rare DR cases would expand generalizability. Finally, the model will be deployed in a clinical setting with real-time feedback and iterative learning to improve its performance further and prove its usefulness in the real world with DR as the target application.

## REFERENCES


[1] Alyoubi, W.L., Shalash, W.M. and Abulkhair, M.F., 2020. Diabetic retinopathy detection through deep learning techniques: A review. Informatics in Medicine Unlocked, 20, p.100377.

[2] Vij, R. and Arora, S., 2023. A systematic review on diabetic retinopathy detection using deep learning techniques. Archives of Computational Methods in Engineering, 30(3), pp.2211-2256.

[3] Bhandari, S., Pathak, S. and Jain, S.A., 2023. A literature review of early-stage diabetic retinopathy detection using deep learning and evolutionary computing techniques. Archives of Computational Methods in Engineering, 30(2), pp.799-810.

[4] Kalyani, G., Janakiramaiah, B., Karuna, A. and Prasad, L.N., 2023. Diabetic retinopathy detection and classification using capsule networks. Complex & Intelligent Systems, 9(3), pp.2651-2664.

[5] Gangwar, A.K. and Ravi, V., 2021. Diabetic retinopathy detection using transfer learning and deep learning. In Evolution in Computational Intelligence: Frontiers in Intelligent Computing: Theory and Applications (FICTA 2020), Volume 1 (pp. 679-689). Springer Singapore.

[6] Kaushik, P. and Khurana, S., 2024, October. OralGuard: Harnessing Inception-ResNet-v2 for Cutting-Edge Oral Health Diagnostics. In 2024 2nd International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS) (pp. 882-887). IEEE.

[7] Muthusamy, D. and Palani, P., 2024. Deep learning model using classification for diabetic retinopathy detection: an overview. Artificial Intelligence Review, 57(7), p.185.

[8] Senapati, A., Tripathy, H.K., Sharma, V. and Gandomi, A.H., 2024. Artificial intelligence for diabetic retinopathy detection: a systematic review. Informatics in Medicine Unlocked, p.101445.

[9] Kaushik, P. and Kaur, S., 2024, October. Advancements in Kidney Disease Diagnosis: The Role of Deep Learning Techniques in Modern Medicine. In 2024 2nd International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS) (pp. 588-592). IEEE.

[10] Luo, X., Wang, W., Xu, Y., Lai, Z., Jin, X., Zhang, B. and Zhang, D., 2024. A deep convolutional neural network for diabetic retinopathy detection via mining local and long‐range dependence. CAAI Transactions on Intelligence Technology, 9(1), pp.153-166.

[11] Das, P.K. and Pumrin, S., 2024. Diabetic Retinopathy Classification: Performance Evaluation of Pre-trained Lightweight CNN using Imbalance Dataset. Engineering Journal, 28(7), pp.13-25.

[12] Ikram, A., Imran, A., Li, J., Alzubaidi, A., Fahim, S., Yasin, A. and Fathi, H., 2024. A systematic review on fundus image-based diabetic retinopathy detection and grading: Current status and future directions. IEEE Access.

[13] Kaushik, P. and Aneja, A., 2024, October. The Future of Orthopedic Care: High-Accuracy Bone Fracture Detection with CNNs. In 2024 2nd International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS) (pp. 193-198). IEEE.

[14] Abushawish, I.Y., Modak, S., Abdel-Raheem, E., Mahmoud, S.A. and Hussain, A.J., 2024. Deep learning in automatic diabetic retinopathy detection and grading systems: a comprehensive survey and comparison of methods. IEEE Access, 12, pp.84785-84802.

[15] https://www.kaggle.com/datasets/sachinkumar413/diabetic-retinopathy-dataset


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